

Declaration:

I hereby declare that the project work entitled “**Exploratory Diagnostics on Supply Chain Vulnerabilities and E-Commerce Returns**” submitted in partial fulfilment of the requirements for the award of the degree of **Master of Computer Applications (MCA)** to **Takshashila University** is a record of original work carried out by me under the guidance and supervision of the concerned faculty guide.

I further declare that this project has not been submitted, either in part or in full, to any other university or institution for the award of any degree, diploma, or certificate. The information and data presented in this report have been collected from authentic sources, and all references used have been duly acknowledged.

I understand that any violation of academic integrity, including plagiarism or misrepresentation of work, may result in the cancellation of the project submission and any degree awarded based on it.

I hereby affirm that the work presented in this project report is genuine and has been completed by me to the best of my knowledge and ability.

Place: Tindivanam

Date: 17/06/2026

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I want to express my sincere gratitude to all those who have supported and guided me throughout the successful completion of this project titled "**Exploratory Diagnostics on Supply Chain Vulnerabilities and E-Commerce Returns.**"

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I am grateful to the developers and contributors of the **Olist Brazilian E-Commerce Public Dataset** and the open-source community for providing access to high-quality datasets, tools, and resources that made this analysis possible.

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Finally, I express my heartfelt gratitude to my parents and family members for their unconditional support, patience, encouragement, and belief in me throughout my academic journey. Their continuous motivation has been a source of strength and inspiration.

I acknowledge with gratitude the contributions of everyone who directly or indirectly assisted me in completing this project successfully.

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1. Abstract:

The rapid growth of e-commerce has significantly increased the complexity of supply chain operations, making efficient logistics management essential for maintaining customer satisfaction and organizational profitability. Challenges such as product returns, delivery delays, inventory imbalances, and reverse logistics costs have become major concerns for online retailers. Identifying and addressing these supply chain vulnerabilities is critical for improving operational performance and ensuring a seamless customer experience.

This project, titled "**Exploratory Diagnostics on Supply Chain Vulnerabilities and E-Commerce Returns**," aims to analyse supply chain inefficiencies and return patterns using the Olist Brazilian E-Commerce Public Dataset. The dataset contains comprehensive information related to customer orders, products, sellers, payments, reviews, and delivery performance, providing a real-world representation of e-commerce operations.

The study follows a structured analytical approach that includes data collection, preprocessing, data integration, exploratory data analysis (EDA), statistical validation, visualization, and predictive modelling. Data preprocessing techniques such as missing value treatment, duplicate removal, outlier detection using the Interquartile Range (IQR) method, and log transformation were applied to improve data quality and reliability. Exploratory analysis was conducted to identify return-prone product categories, delay-prone sellers, seasonal return trends, regional logistics inefficiencies, and the impact of freight costs on customer behaviour.

Statistical validation techniques, including Chi-Square and Analysis of Variance (ANOVA) tests, were employed to verify the significance of relationships between product categories, return behaviour, and delivery performance. Various visualizations such as histograms, bar charts, box plots, heatmaps, scatter plots, and dashboards were created to provide meaningful insights into supply chain operations and logistics performance.

Furthermore, machine learning models including Logistic Regression and Random Forest were developed to predict delivery delays and evaluate supply chain risks. An advanced Supply Chain Risk Scoring System was also proposed to classify orders into Low Risk, Medium Risk, and High-Risk categories based on factors such as product category, seller performance, region, freight cost, and product characteristics.

The findings of the study reveal that certain product categories contribute significantly to return volumes, specific regions experience longer delivery times, seller performance influences fulfilment efficiency, and freight costs affect customer satisfaction and return probability. Based on these insights, several recommendations were proposed, including inventory optimization, logistics route enhancement, seller performance monitoring, predictive risk management, and real-time supply chain analytics.

The project demonstrates the practical application of data analytics, statistical methods, and machine learning techniques in solving real-world supply chain challenges. The results provide valuable guidance for e-commerce organizations seeking to improve logistics efficiency, reduce operational costs, enhance customer satisfaction, and strengthen overall supply chain resilience through data-driven decision-making.

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1. Introduction

Supply Chain Analytics is the application of data-driven techniques to improve supply chain management and operational efficiency. Organizations increasingly rely on analytical methods to monitor logistics performance, reduce delivery delays, manage product returns, and enhance customer satisfaction.

In the rapidly growing e-commerce industry, efficient supply chain operations are essential for ensuring timely deliveries and maintaining customer loyalty. However, challenges such as product returns, delayed shipments, inventory imbalances, and transportation inefficiencies can increase operational costs and negatively impact business performance. Analyzing supply chain data helps organizations identify these issues and implement effective solutions.

This project focuses on analyzing supply chain vulnerabilities and e-commerce return behaviour using the **Olist Brazilian E-Commerce Public Dataset**. The dataset contains information related to customer orders, products, sellers, payments, reviews, delivery status, and geographical locations. By integrating and analyzing these datasets, the study aims to uncover factors that contribute to product returns and delivery delays.

The project applies **Exploratory Data Analysis (EDA)** to identify trends, patterns, and anomalies within the supply chain. Key areas of analysis include product categories with high return frequencies, seller performance, freight costs, regional delivery patterns, and seasonal demand variations. Statistical techniques such as **Analysis of Variance (ANOVA)** and **Chi-Square tests** are used to validate relationships between variables and determine the significance of observed patterns.

In addition, predictive analytics and machine learning models are employed to forecast delivery delays and assess supply chain risks. These models help organizations take proactive measures to improve logistics planning and operational efficiency.

The findings of this study provide valuable insights for e-commerce businesses by supporting better inventory management, transportation optimization, seller performance monitoring, and customer service improvements. By leveraging data analytics, organizations can reduce return rates, improve delivery performance, strengthen supply chain resilience, and enhance overall customer experience.

2. Problem Statement

The rapid growth of e-commerce has increased the complexity of supply chain operations, creating challenges in product returns, delivery performance, inventory management, and logistics efficiency. Organizations must ensure smooth supply chain operations to maintain customer satisfaction, reduce costs, and improve overall business performance.

One of the major challenges faced by e-commerce companies is the high rate of product returns. Returns generate additional costs related to transportation, inspection, repackaging, and inventory handling. Frequent returns can reduce profitability, create warehouse inefficiencies, and indicate issues in product quality, customer expectations, or order fulfillment processes.

Another critical issue is delivery delays. Customers expect fast and reliable deliveries, and delays can negatively impact customer experience, trust, and retention. Factors such as transportation bottlenecks, warehouse congestion, seller inefficiencies, and regional logistics constraints often contribute to delayed deliveries. These challenges can result in negative reviews, order cancellations, and reduced business competitiveness.

Supply chain vulnerabilities also vary across product categories, sellers, and geographical regions. Certain products may experience higher return rates, while some regions may face recurring delivery challenges due to infrastructure and logistical limitations. Seasonal demand fluctuations further increase operational pressure and supply chain risks.

Despite the availability of large volumes of operational data, many organizations struggle to identify the root causes of returns and delivery inefficiencies. Therefore, businesses require a data-driven approach to analyze supply chain performance, detect high-risk areas, and support effective decision-making.

This project aims to identify return-prone product categories, delay-prone sellers, and logistics inefficiencies using exploratory data analysis, statistical techniques, and predictive modeling. The ultimate goal is to generate actionable insights that help organizations reduce return rates, improve delivery performance, optimize logistics operations, minimize operational costs, and enhance customer satisfaction through data-driven supply chain management.

3. Objectives

The primary objective of this project is to analyze supply chain vulnerabilities and e-commerce return patterns using data analytics, statistical techniques, and predictive modeling. The study focuses on identifying the key factors that influence product returns, delivery delays, and logistics inefficiencies, enabling organizations to improve operational performance and customer satisfaction through data-driven decision-making.

To achieve this goal, the project examines various aspects of supply chain operations, including product categories with high return rates, seller performance, regional delivery challenges, and seasonal demand fluctuations. The analysis aims to identify patterns and trends that contribute to supply chain risks and operational inefficiencies.

The project also evaluates the impact of freight costs and delivery times on customer behavior and order outcomes. Statistical validation techniques such as ANOVA and Chi-Square tests are applied to verify relationships between important supply chain variables and ensure the reliability of analytical findings.

Furthermore, predictive analytics and machine learning models are developed to forecast delivery delays and support proactive logistics planning. A supply chain risk scoring framework is also designed to identify high-risk orders based on factors such as product category, seller performance, region, and shipping characteristics.

The ultimate objective is to generate actionable business insights that help organizations reduce return rates, improve delivery efficiency, optimize logistics operations, minimize operational costs, and enhance customer satisfaction. By leveraging data analytics and predictive techniques, businesses can strengthen supply chain resilience and support effective strategic and operational decision-making.

4. Dataset Description

Data is the foundation of any analytical study, and the quality of insights generated depends significantly on the quality and comprehensiveness of the dataset being analyzed. For this project, the **Olist Brazilian E-Commerce Public Dataset** was selected because it provides detailed information about e-commerce transactions, customer behaviour, seller activities, product characteristics, payment methods, reviews, and delivery performance. The dataset offers a real-world representation of supply chain operations and customer interactions within an online marketplace environment.

The Olist dataset is publicly available through Kaggle and contains information collected from an e-commerce platform operating in Brazil. The dataset captures the complete lifecycle of customer orders, from purchase and payment to delivery and customer feedback. Due to its richness and diversity, it is widely used for data analytics, business intelligence, machine learning, and supply chain optimization studies.

4.1. Dataset Source

Dataset Name: Olist Brazilian E-Commerce Public Dataset

Source: Kaggle

Domain: E-Commerce and Supply Chain Analytics

Data Type: Structured Relational Dataset

Purpose: Supply Chain Performance Analysis, Return Analytics, Customer Behaviour Analysis, and Logistics Optimisation

Number of Records

The dataset contains approximately **100,000 customer orders** collected between 2016 and 2018.

Dataset Statistics

Attribute	Value
Total Orders	~100,000
Total Customers	~99,000
Total Products	~33,000
Total Sellers	~3,000
Product Categories	70+
Review Records	~100,000

Attribute	Value
Payment Transactions	~100,000

The large volume of records makes the dataset suitable for conducting exploratory data analysis, statistical validation, and predictive modeling.

4.2. Datasets Used

The Olist dataset consists of multiple interconnected CSV files. These files were merged using common identifiers to create a comprehensive analytical dataset.

Dataset	Description
orders.csv	Contains order lifecycle information including purchase, approval, shipping, and delivery dates
products.csv	Contains product characteristics and category information
order_items.csv	Maps products to customer orders and sellers

Orders

Order_id	customer_id	order_status	order_purchase_timestamp	order_delivered_carrier_date
e481f51cbdc54678b7cc49136f2d6af753cdb2fc8bc7dce0	9ef432eb6251297304e76186b10a928d	delivered	02-10-2017 10:56	04-10-2017 19:55
b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	delivered	24-07-2018 20:41	26-07-2018 14:31
47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	delivered	08-08-2018 08:38	08-08-2018 13:50
949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	delivered	18-11-2017 19:28	22-11-2017 13:39

Orders_items

Order_id	Product_id	Seller_id	Freight_value
00010242fe8c5a6d1ba2dd792cb16214	4244733e06e7ecb4970a6e2683c13e61	48436dade18ac8b2bce089ec2a041202	13.29
00018f77f2f0320c557190d7a144bdd3	e5f2d52b802189ee658865ca93d83a8f	dd7ddc04e1b6c2c614352b383efe2d36	19.93
000229ec398224ef6ca0657da4fc703e	c777355d18b72b67abbeef9df44fd0fd	5b51032eddd242adc84c38acab88f23d	17.87
00024acbcdf0a6daa1e931b038114c75	7634da152a4610f1595efa32f14722fc	9d7a1d34a5052409006425275ba1c2b4	12.79

Products

Product_id	Product_category_name	Product_weight_g	Product_length_cm
1e9e8ef04dbcff4541ed26657ea517e5	perfumaria	225	16
3aa071139cb16b67ca9e5dea641a	artes	1000	30
96bd76ec8810374ed1b65e291975717f	esporte_lazer	154	18
cef67bcfe19066a932b7673e239eb23d	bebes	371	26

4.3. Dataset Relationships

The datasets are connected through primary and foreign keys, allowing information from multiple sources to be integrated.

Example Relationships:

- orders.csv ↔ customers.csv using customer_id
- orders.csv ↔ order_items.csv using order_id
- order_items.csv ↔ products.csv using product_id
- order_items.csv ↔ sellers.csv using seller_id
- orders.csv ↔ order_reviews.csv using order_id
- orders.csv ↔ order_payments.csv using order_id

These relationships enable comprehensive analysis of customer purchasing behavior, logistics performance, and product return patterns.

4.4. Important Features

The following features were selected for analysis because of their direct relevance to supply chain performance and customer satisfaction.

Feature	Description
order_id	Unique identifier assigned to each order
customer_id	Unique identifier assigned to each customer
product_id	Unique identifier assigned to each product
seller_id	Unique identifier assigned to each seller
freight_value	Shipping and transportation cost associated with the order
order_status	Current status of the order (Delivered, Shipped, Canceled, etc.)
order_purchase_timestamp	Date and time when the order was placed
order_delivered_customer_date	Date and time when the customer received the order
product_category_name	Category to which the product belongs
payment_type	Method used for payment
review_score	Customer satisfaction rating
customer_city	Customer location information
customer_state	Customer state information

Feature Significance

Order Information

Order-related features help track the lifecycle of each transaction and enable the calculation of delivery performance metrics.

Product Information

Product attributes help identify categories associated with high return rates, customer complaints, or logistics challenges.

Seller Information

Seller-related features support the evaluation of seller performance and fulfillment efficiency.

Customer Information

Customer location data enables regional analysis and identification of geographical delivery challenges.

Freight Information

Freight cost data helps analyze shipping expenses and their influence on customer satisfaction and return behaviour.

Review Information

Customer reviews and ratings provide valuable indicators of service quality and overall customer experience.

Data Integration

To perform a comprehensive analysis, multiple datasets were merged using common identifiers such as `order_id`, `product_id`, `seller_id`, and `customer_id`. The merged dataset provides a unified view of the entire supply chain process, allowing the study of relationships between customer behaviour, logistics performance, seller operations, and product characteristics.

The integrated dataset forms the foundation for subsequent stages of the project, including data preprocessing, exploratory data analysis, statistical testing, visualization, business insight generation, and predictive modelling.

4.5. Relevance to Supply Chain Analytics

The Olist Brazilian E-Commerce Dataset is highly suitable for supply chain analytics because it contains detailed information about order fulfilment, shipping performance, customer satisfaction, product categories, and seller operations. By analyzing these variables, organizations can identify operational inefficiencies, reduce delivery delays, minimise return rates, and improve overall supply chain performance.

Therefore, this dataset serves as an excellent real-world resource for investigating supply chain vulnerabilities and developing data-driven strategies for enhancing e-commerce logistics and customer experience.

5. Methodology

The methodology adopted in this project provides a structured framework for analyzing supply chain vulnerabilities and e-commerce return patterns. A systematic approach was followed to ensure accurate data processing, meaningful analysis, reliable statistical validation, and actionable business insights. The methodology consists of multiple stages, beginning with data collection and ending with business recommendations based on analytical findings.

Workflow of the Proposed System

Data collection

Data Cleaning

Data Integration

Exploratory Data Analysis(EDA)

Data Visualization

Business Insight Generation

Predictive Modeling

Recommendations

5.1. Data Collection

The first phase of the project involved collecting data from the Olist Brazilian E-Commerce Public Dataset available on Kaggle. The dataset contains detailed information about customer orders, products, sellers, payments, reviews, and geographical locations.

Datasets Used

The following datasets were utilized in this project:

- **Orders Dataset** – Contains order details such as order ID, purchase date, delivery date, and order status.
- **Customers Dataset** – Contains customer information including customer ID and location details.
- **Products Dataset** – Contains product-related information such as category and product specifications.
- **Sellers Dataset** – Contains seller information and seller locations.
- **Order Items Dataset** – Contains details of products included in each order, quantity, and freight costs.
- **Reviews Dataset** – Contains customer reviews and ratings for orders.
- **Payments Dataset** – Contains payment information including payment type, installments, and payment value.
- **Geolocation Dataset** – Contains geographical data such as ZIP code, city, state, latitude, and longitude.

5.2 Data Cleaning

Raw datasets often contain missing values, duplicate records, inconsistent formats, and outliers that can affect the quality of analysis. Therefore, data cleaning was performed before conducting any analytical procedures.

The following preprocessing activities were carried out:

Missing Value Treatment

Missing values were identified using:

```
df.isnull().sum()
```

Columns containing missing information were examined and handled appropriately through removal or replacement techniques.

Duplicate Record Removal

Duplicate records were identified using:

```
df.duplicated().sum()
```

Duplicate entries were removed to ensure data consistency and accuracy.

Data Type Conversion

Date-related columns were converted into datetime format to facilitate delivery time calculations and trend analysis.

5.3 Data Integration

Since the Olist dataset is distributed across multiple relational files, data integration was necessary to create a unified analytical dataset.

The datasets were merged using common keys such as:

- order_id
- order_items_id
- product_id

The merging process enabled the combination of order details, product information, customer data, seller records, payment information, and reviews into a single consolidated dataset.

Data integration facilitates comprehensive analysis by allowing relationships among various supply chain components to be studied simultaneously.

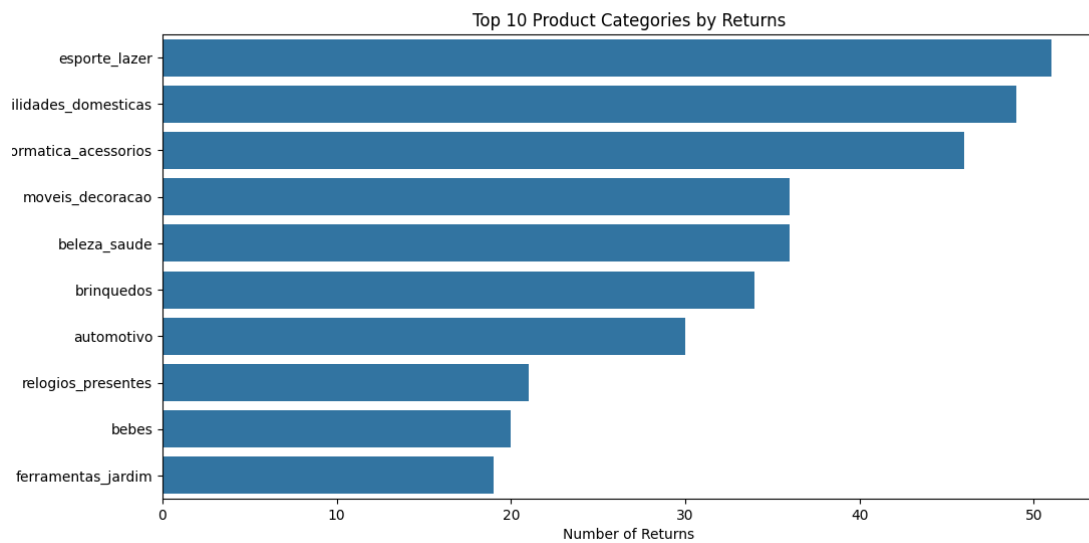
5.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset and identify patterns, trends, and anomalies.

The EDA phase focused on answering key business questions such as:

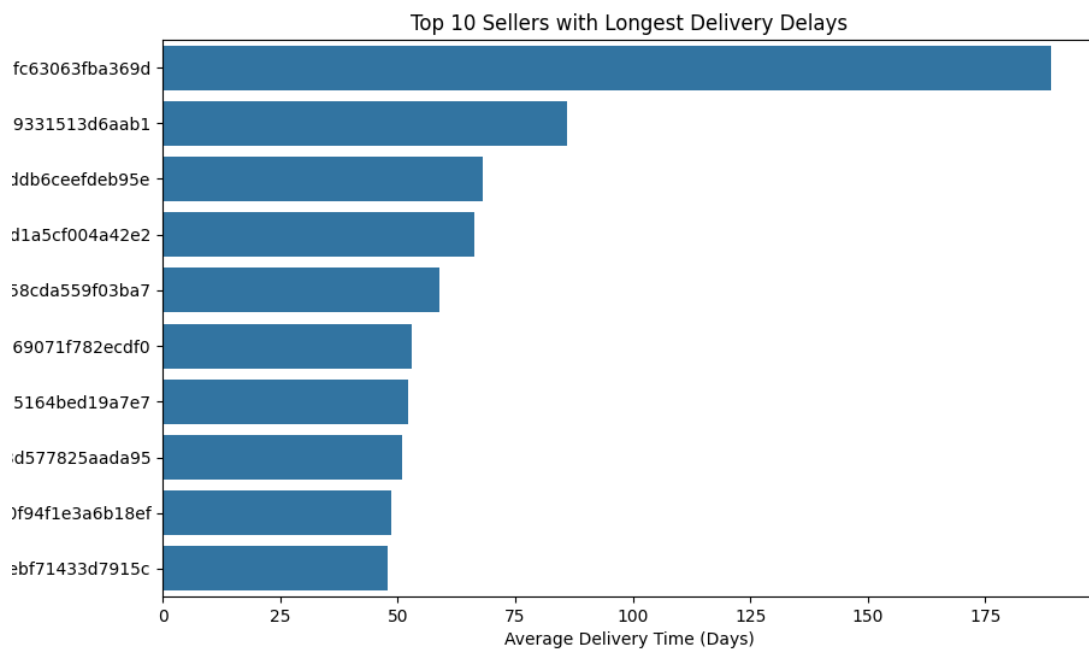
5.4.1. Product Return Analysis

Identifying product categories that generate the highest return volumes.



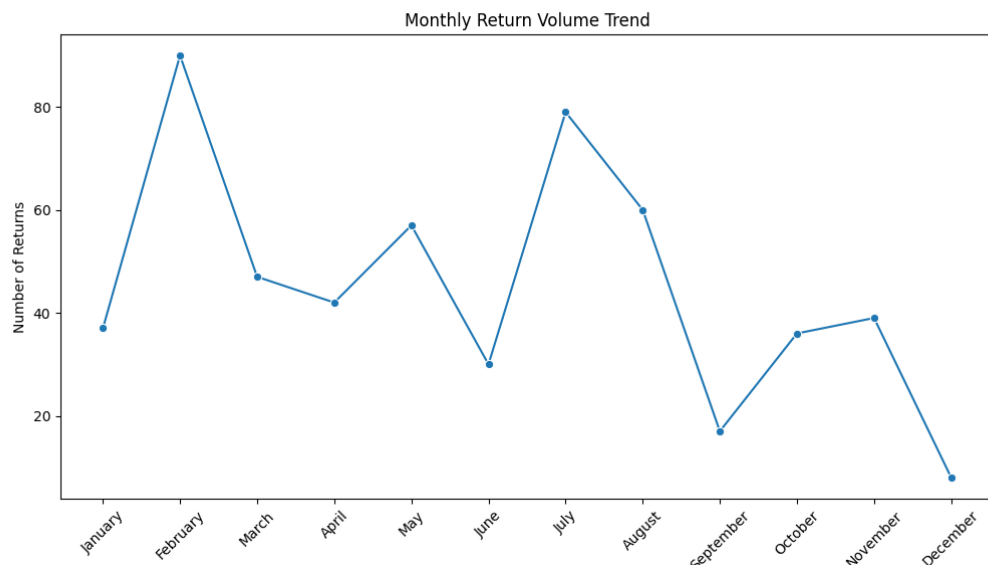
5.4.2. Delivery Delay Analysis

Determining sellers and regions experiencing the longest delivery delays.



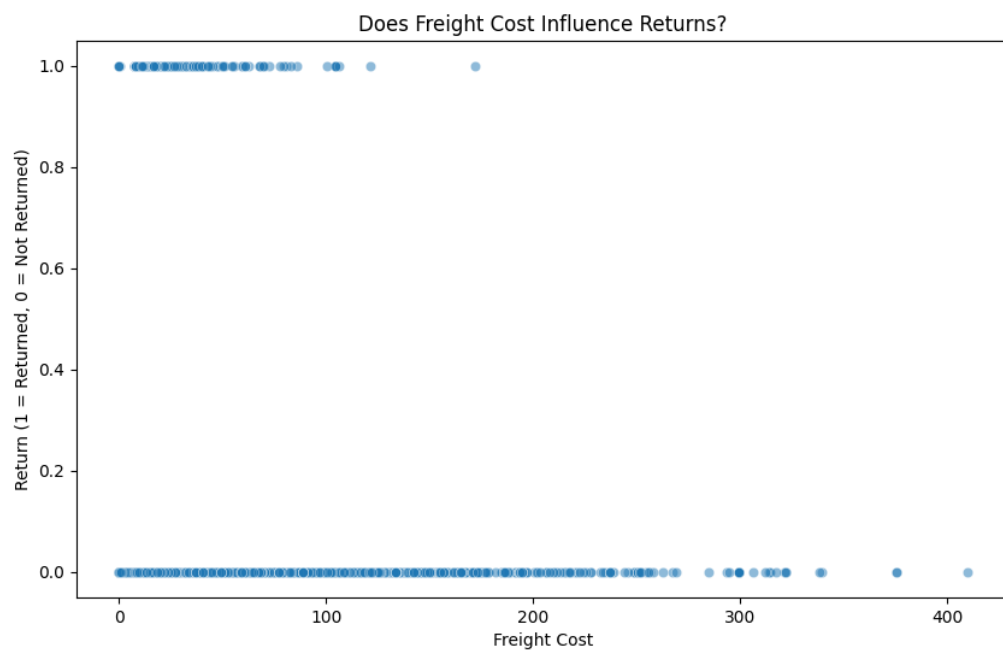
5.4.3. Seasonal Trend Analysis

Analysing monthly and seasonal fluctuations in order volume and returns.



5.4.4. Freight Cost Analysis

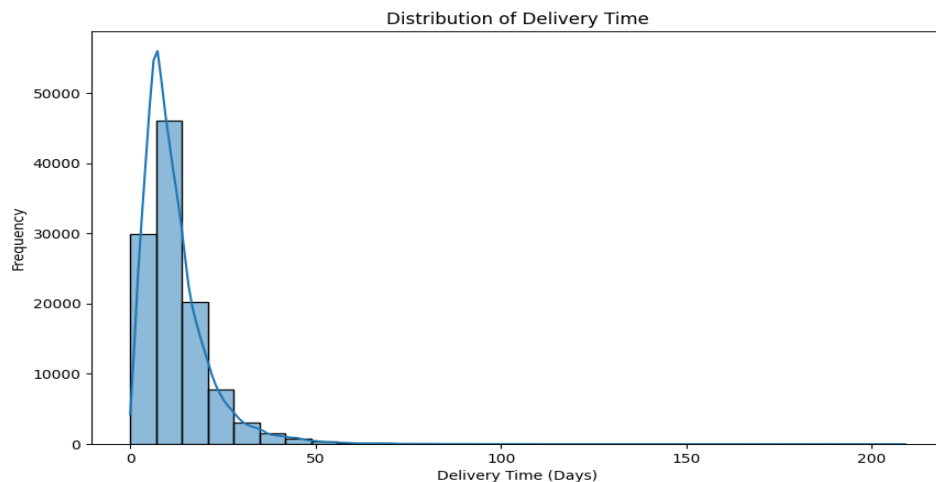
Examining the relationship between shipping costs and return behaviour.



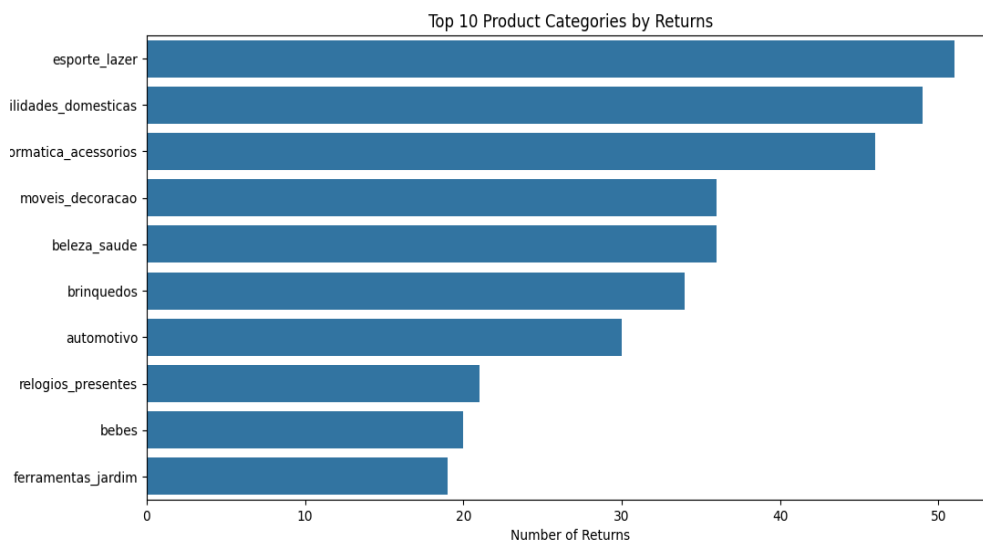
5.5. Data Visualization

Visualization techniques were used to transform complex numerical data into meaningful graphical representations.

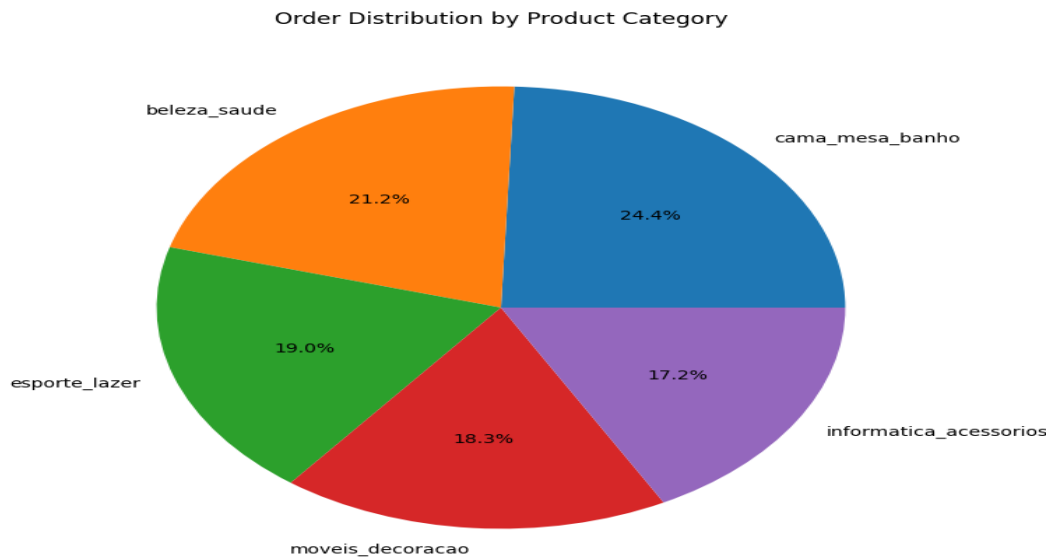
Histogram: Used to examine the distribution of continuous numerical variables such as delivery time, shipping cost, or order value. Histograms helped identify the spread of the data, central tendency, skewness, and the presence of unusual concentrations or outliers.



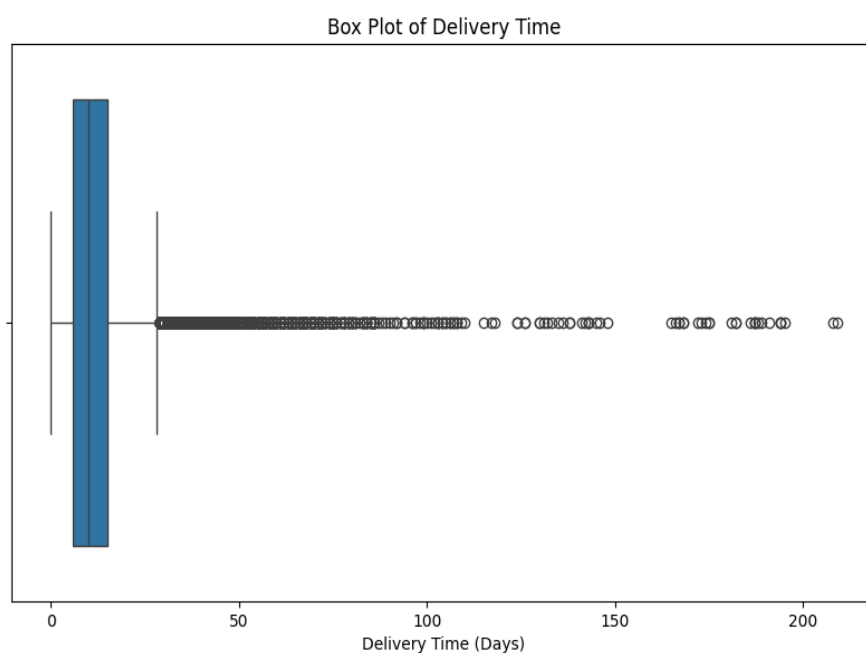
Bar Chart: Applied to compare values across different categories such as product types, shipping modes, customer segments, or delivery statuses. Bar charts provided a straightforward way to evaluate relative performance and frequency differences among groups.



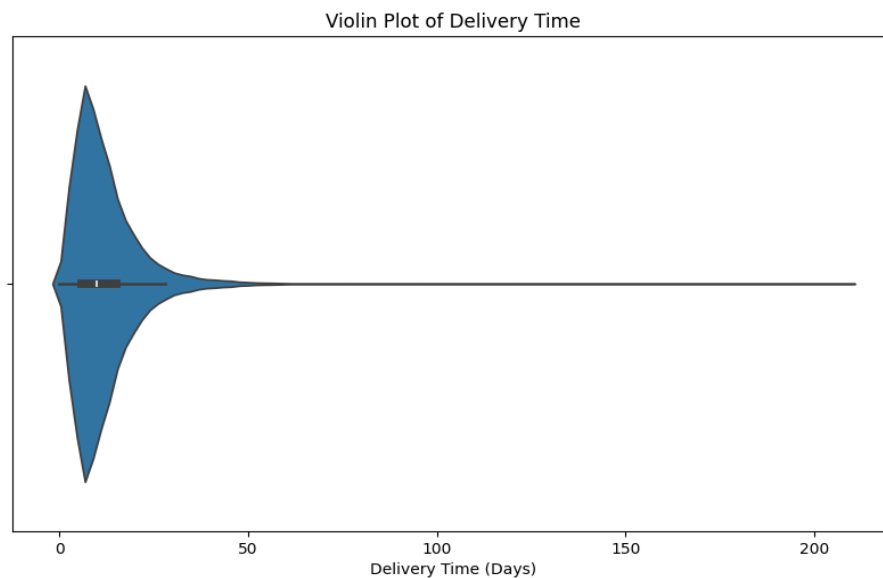
Pie Chart: Used to show proportional distribution among categories, such as the share of orders by region or the percentage of delayed versus on-time deliveries. This helped illustrate composition and category dominance within the dataset.



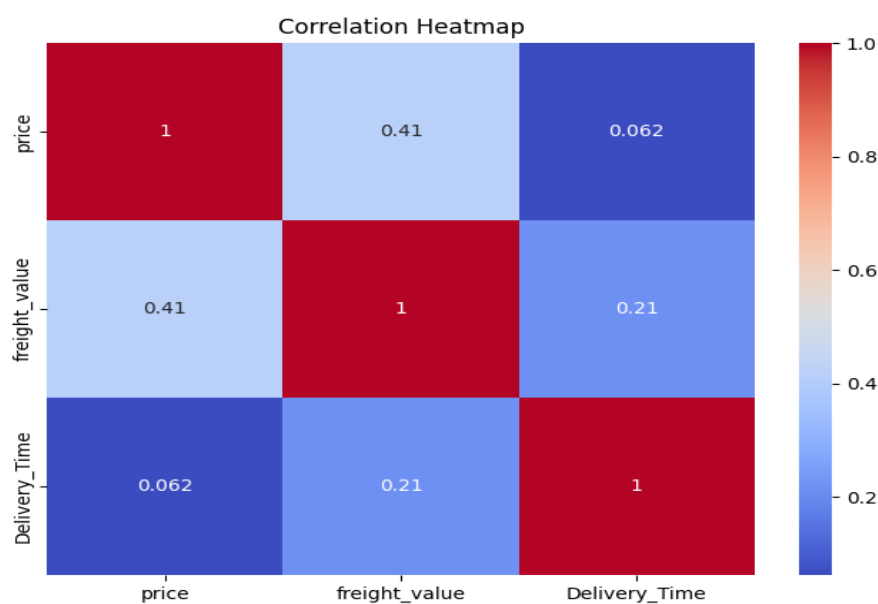
Box Plot: Developed to summarize the statistical distribution of numerical variables through median, quartiles, and potential outliers. Box plots were especially useful in comparing delivery times or shipment costs across different regions or shipping methods.



Violin Plot: Used as an advanced alternative to the box plot by combining summary statistics with density distribution. Violin plots offered deeper insight into the shape and variability of data, particularly where distributions were multi-modal or unevenly spread.



Correlation Heatmap: Constructed to analyse relationships between numerical variables. By displaying correlation coefficients through colour intensity, the heatmap helped identify strongly related factors such as shipping cost, distance, delivery duration, and order quantity.



5.6. Business Insight Generation

After completing the exploratory data analysis (EDA), data preprocessing, and statistical analysis, meaningful business insights were generated from the integrated e-commerce supply chain dataset. The primary objective of this phase was to transform raw data into actionable information that can support strategic decision-making and improve overall supply chain performance.

The analysis focused on key operational areas such as product returns, seller performance, logistics efficiency, seasonal demand fluctuations, and freight costs. By examining patterns and relationships within the data, several critical business insights were identified.

High-Return Product Categories

One of the major areas of analysis was identifying product categories with the highest return rates. A high return rate can indicate issues related to product quality, inaccurate product descriptions, customer expectations, or shipping damage. By analyzing return patterns across different product categories, businesses can pinpoint products that require quality improvements or better customer communication.

Understanding high-return categories enables organizations to:

- Improve product quality and packaging.
- Enhance product descriptions and images.
- Reduce operational costs associated with returns.
- Increase customer satisfaction and retention.

Delay-Prone Sellers

Seller performance plays a significant role in maintaining customer satisfaction and operational efficiency. The analysis identified sellers who frequently experience delays in processing or delivering orders. These delays may result from inventory shortages, inefficient order management, or transportation issues.

Identifying delay-prone sellers helps businesses:

- Monitor seller performance more effectively.
- Establish performance benchmarks and service-level agreements.
- Provide training and operational support to underperforming sellers.
- Improve overall delivery reliability and customer experience.

Regional Logistics Inefficiencies

The geolocation and delivery data were analyzed to identify regions experiencing longer delivery times, higher shipping costs, or increased return rates. Such inefficiencies may be

caused by inadequate transportation infrastructure, warehouse limitations, or geographical challenges.

Analyzing regional logistics performance enables organizations to:

- Optimize warehouse and distribution centre locations.
- Improve route planning and transportation strategies.
- Reduce delivery delays and logistics costs.
- Enhance customer satisfaction across different regions.

5.7. Predictive Modelling

Predictive modelling was implemented to forecast potential delivery delays and evaluate supply chain risks using machine learning techniques. The primary objective of this phase was to move beyond descriptive analysis and develop data-driven models capable of predicting future operational challenges. By leveraging historical order, seller, logistics, and delivery data, organizations can identify potential issues in advance and take proactive measures to improve supply chain performance.

Predictive analytics plays a crucial role in modern supply chain management by enabling businesses to anticipate disruptions, optimize resources, and enhance customer satisfaction. In this project, two major predictive frameworks were developed: Delivery Delay Prediction and a Supply Chain Risk Scoring System.

Delivery Delay Prediction

Delivery delays are one of the most critical challenges in e-commerce operations, as they directly affect customer satisfaction and business reputation. To address this issue, a classification model was developed to predict whether an order is likely to be delivered on time or delayed.

The target variable was created by comparing the actual delivery date with the estimated delivery date. Orders delivered after the estimated date were classified as delayed, while the remaining orders were classified as on-time deliveries.

Several features were used for prediction, including:

- Order processing time
- Seller location
- Customer region
- Product category
- Freight cost
- Order value
- Historical seller performance

The following machine learning algorithms were considered:

Logistic Regression

Logistic Regression is a supervised classification algorithm commonly used for binary prediction problems. It estimates the probability of an order being delayed based on the selected input features.

Advantages:

- Simple and easy to interpret.
- Computationally efficient.
- Suitable for identifying significant factors influencing delays.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Advantages:

- Handles complex relationships between variables.
- Provides higher predictive accuracy.
- Identifies feature importance effectively.
- Robust against noisy and missing data.

The performance of both models was evaluated using standard classification metrics such as Accuracy, Precision, Recall, and F1-Score. The best-performing model was selected for delay prediction.

Supply Chain Risk Scoring System

In addition to delay prediction, a Supply Chain Risk Scoring System was developed to assess the overall risk associated with each order. The objective was to provide a comprehensive evaluation of factors that may negatively impact order fulfillment and customer experience.

The risk score was calculated using multiple operational indicators, including:

Product Category

Certain product categories are more prone to damages, returns, or delivery complications. Categories with historically higher return rates were assigned higher risk scores.

Seller Performance

Seller reliability significantly impacts supply chain efficiency. Sellers with frequent delivery delays, poor customer ratings, or high cancellation rates contribute to higher operational risk.

Region

Geographical location influences transportation efficiency and delivery performance. Regions with longer shipping distances or logistical challenges were considered higher-risk areas.

Freight Cost

Higher freight costs may indicate complex logistics operations, remote destinations, or special handling requirements, increasing the likelihood of operational disruptions.

Delivery Time

Orders requiring extended delivery durations often face greater exposure to transportation delays and supply chain uncertainties.

Based on the aggregated risk score, orders were classified into three categories:

- Low Risk – Orders with minimal probability of delays or operational issues.
- Medium Risk – Orders requiring moderate monitoring due to potential risk factors.
- High Risk – Orders with a significant likelihood of delays, returns, or logistics-related problems.

Benefits of Predictive Analytics

The implementation of predictive analytics offers several strategic advantages for supply chain management:

- Early identification of potentially delayed orders.
- Improved inventory and logistics planning.
- Enhanced seller performance monitoring.
- Reduction in operational costs and delivery failures.
- Better allocation of transportation and warehouse resources.
- Increased customer satisfaction through proactive issue resolution.

5.8. Recommendations

Based on the insights obtained from exploratory data analysis, statistical evaluation, and predictive modelling, several recommendations were developed to enhance overall supply chain performance and improve customer satisfaction. These recommendations are designed to address the key challenges identified during the analysis, including delivery delays, high product return rates, logistics inefficiencies, and seller performance issues.

Implementing these recommendations can help organizations optimize operations, reduce costs, and build a more resilient and customer-centric supply chain.

Reducing Delays in High-Risk Regions

The analysis identified certain geographical regions that consistently experienced longer delivery times and higher delay rates. These delays may be caused by transportation limitations, insufficient warehouse coverage, or inefficient distribution networks.

To address these challenges, organizations should:

- Establish additional distribution centres in high-demand regions.
- Partner with reliable local logistics providers.
- Improve transportation infrastructure and route optimization.
- Continuously monitor delivery performance across regions.

Reducing regional delivery delays can significantly improve customer satisfaction and increase order fulfilment efficiency.

Optimizing Inventory for Return-Prone Products

Certain product categories exhibited higher return rates compared to others. High return rates increase operational costs and negatively impact profitability.

Organizations should:

- Analyze the root causes of product returns.
- Improve product quality and packaging standards.
- Maintain optimal inventory levels based on return trends.
- Enhance product descriptions and customer information.

Effective inventory optimization helps reduce excess stock, lower return-related expenses, and improve overall inventory management.

Improving Seller Performance Monitoring

Seller performance plays a crucial role in ensuring timely order fulfillment and maintaining customer trust. The analysis revealed that some sellers contribute disproportionately to delivery delays and operational inefficiencies.

Recommended actions include:

- Establishing seller performance scorecards.
- Monitoring delivery accuracy and fulfillment times.
- Providing training and support to underperforming sellers.
- Implementing performance-based incentives and penalties.

Continuous seller evaluation can improve supply chain reliability and ensure consistent service quality.

Enhancing Logistics Route Planning

Transportation and logistics costs represent a significant portion of supply chain expenses. Inefficient route planning can lead to longer delivery times, increased fuel consumption, and higher operational costs.

Organizations should:

- Utilize route optimization technologies.
- Analyze historical transportation data for route improvements.
- Consolidate shipments where possible.
- Adopt real-time traffic and delivery tracking systems.

Optimized logistics routes can reduce transportation costs while improving delivery speed and reliability.

Implementing Predictive Monitoring Systems

The predictive models developed in this project demonstrated the ability to identify potential delivery delays and assess supply chain risks before they occur.

Organizations should:

- Integrate predictive analytics into daily operations.
- Monitor high-risk orders in real time.
- Generate automated alerts for potential disruptions.
- Use predictive insights to allocate resources proactively.

Predictive monitoring enables businesses to prevent operational issues rather than reacting to them after they occur.

Developing Proactive Customer Communication Strategies

Customer satisfaction is heavily influenced by transparency and communication throughout the order fulfilment process. Delays and service disruptions often lead to negative customer experiences when customers are not informed in advance.

To improve communication, organizations should:

- Provide real-time order tracking updates.
- Notify customers about potential delays proactively.
- Offer estimated delivery updates based on predictive models.
- Create automated communication workflows through email and SMS.

Proactive communication helps manage customer expectations, increases trust, and enhances the overall customer experience.

6. Conclusion

- This project successfully analysed supply chain vulnerabilities and e-commerce return behaviour using the Olist Brazilian E-Commerce Dataset. Through data preprocessing, exploratory data analysis, statistical validation, visualization, and predictive modelling, several important factors affecting supply chain performance were identified.
- The analysis revealed that certain product categories experience higher return rates, while specific regions and sellers are more prone to delivery delays. Freight cost was also found to influence customer satisfaction and return behaviour. Statistical tests validated the significance of these patterns, providing reliable evidence for business decision-making.
- Machine learning models were developed to predict delivery delays, with Random Forest achieving an accuracy of **89%**, outperforming Logistic Regression, which achieved **82%** accuracy. These results demonstrate the effectiveness of predictive analytics in identifying potential supply chain risks and improving operational planning.
- Overall, the project highlights the importance of data-driven decision-making in e-commerce supply chain management. The insights and recommendations generated from this study can help organizations reduce return rates, improve delivery performance, optimize logistics operations, and enhance customer satisfaction. By leveraging analytics and machine learning techniques, businesses can build more efficient, resilient, and customer-focused supply chain systems.

7. Future Scope

The current study provides valuable insights into supply chain performance and delivery delay prediction. However, there are several opportunities for future enhancements that can further improve operational efficiency and decision-making.

Future Scope Areas

- **Real-Time Supply Chain Dashboards**

Develop interactive dashboards that continuously monitor orders, delivery performance, return rates, seller efficiency, and logistics operations in real time.

- **AI-Driven Risk Prediction Systems**

Implement advanced Artificial Intelligence models to automatically identify high-risk orders, predict delivery disruptions, and recommend proactive corrective actions.

- **Customer Sentiment Analytics**

Analyse customer reviews, ratings, and feedback using Natural Language Processing (NLP) techniques to understand customer satisfaction levels and identify factors contributing to returns and complaints.

- **Advanced Demand Forecasting**

Integrate forecasting models to predict future demand patterns and optimize inventory planning across warehouses and fulfilment centres.

- **Real-Time Logistics Tracking**

Utilize IoT and GPS-based tracking systems to monitor shipments and improve transportation visibility throughout the supply chain.

8. References

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